CS5801 Coursework Launch

Quantitative Data Analysis — Assessment: Coursework

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2026-10-28

Before we start



This session is being recorded

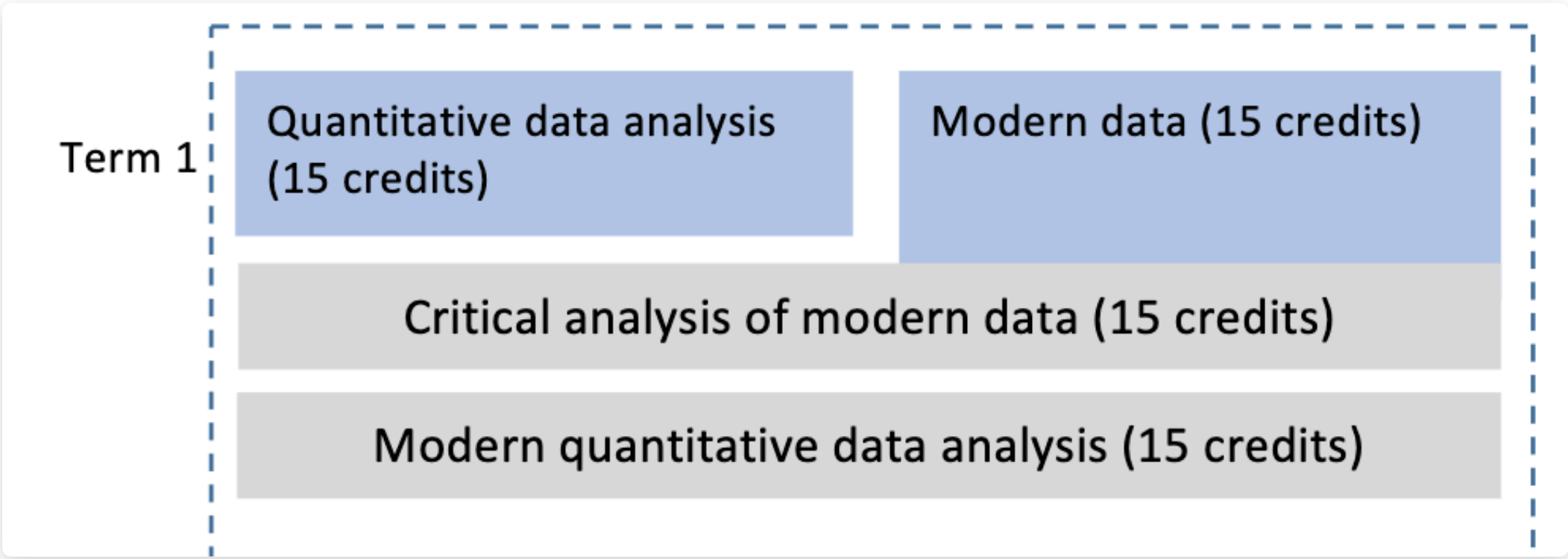


> **Note**
>
> The definitive version of the assessment is the written assessment brief document available on Brightspace.

About CS5801

- Is an **individual** coursework assessment
- The assessment **brief** is on the Brightspace CS5801 page
- The coursework is due by **11:00 am** on **04/01/2027**
- You must submit to **WiseFlow**
- This is a **shared** assessment block for CS5701 and CS5702
- There is a second shared assessment block (CS5802) that is a closed book examination

Assessment and Teaching Blocks



TL;DR

- **Individual** coursework — due **4 January 2027, 11:00 (UK time)**, submitted via **WiseFlow**
- Build **your own dataset** using your student ID (template provided) — your analysis will differ from everyone else's
- Four tasks: clean & explore the data → model a **numeric** outcome → model a **binary** outcome (use `step()` — there are many candidate predictors)
- Submit **two files**: `*-submission-26-27.Rmd` (this is what's marked) and `*-supporting-26-27.Rmd` (your working/evidence)
- **1600-word limit** — be specific to *your* dataset, not generic
- GenAI use is fine **with disclosure**; the work submitted is your own
- **Start early**, ask early — we won't debug your code or pre-mark, but we will help

Planning your work

- You can start working on this now but we have *not* covered all the material required to complete the whole coursework yet
- Don’t leave this to the last minute - Brunel will be closed from 14/12 -> 1/1 (Friday)
- The submission deadline is the 4th of January 11:00 am uk time on Wiseflow
- You can submit sooner, if you wish
- If you need support **do not leave it till then**
- See the assessment brief on Brightspace for late submission penalties

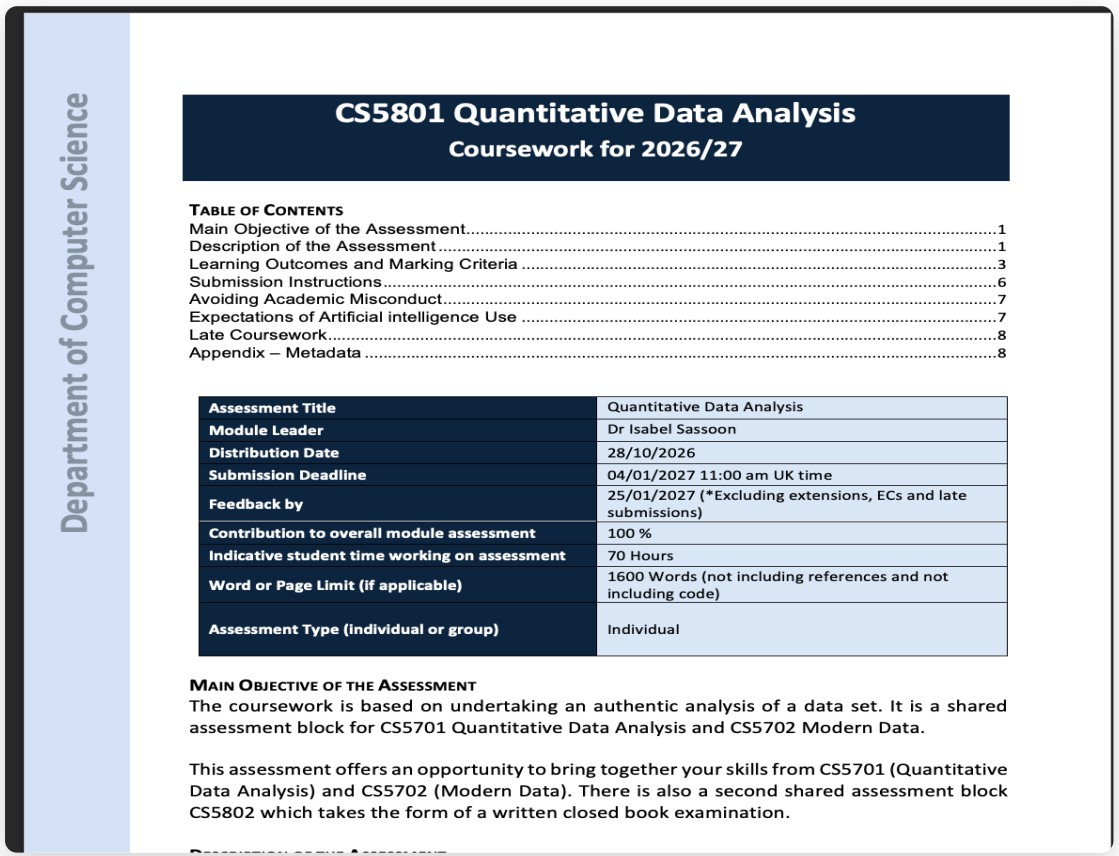
Coursework - description

You will be provided with:

- The Assessment Brief
- A data set and its metadata (**Health club**)
- A set of questions to address
- 2 pro forma (Rmd templates) to use to complete it
- Each student must create (`subset()`) an individual data set (code provided)
- Working with realistic data means that you could end up with models that do not have good fit – if that is the case you will not be **penalised**

Assessment Brief

Available for download from CS5801 Brightspace



Department of Computer Science

CS5801 Quantitative Data Analysis
Coursework for 2026/27

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Assessment Title	Quantitative Data Analysis
Module Leader	Dr Isabel Sassoon
Distribution Date	28/10/2026
Submission Deadline	04/01/2027 11:00 am UK time
Feedback by	25/01/2027 (*Excluding extensions, ECs and late submissions)
Contribution to overall module assessment	100 %
Indicative student time working on assessment	70 Hours
Word or Page Limit (if applicable)	1600 Words (not including references and not including code)
Assessment Type (individual or group)	Individual

MAIN OBJECTIVE OF THE ASSESSMENT

The coursework is based on undertaking an authentic analysis of a data set. It is a shared assessment block for CS5701 Quantitative Data Analysis and CS5702 Modern Data.

This assessment offers an opportunity to bring together your skills from CS5701 (Quantitative Data Analysis) and CS5702 (Modern Data). There is also a second shared assessment block CS5802 which takes the form of a written closed book examination.

Coursework - sections

1. Organise and clean the data
2. Exploratory data analysis
3. Modelling the variable: **monthly spend**
4. Modelling likelihood of **cancelled**

Coursework - individual data

- You must create your own data set from the master data set **based on your student id**
- Use the **template provided** to extract the necessary data (same in both `rmd` files provided)
- This means your analyses and models will **differ**!

Individual data

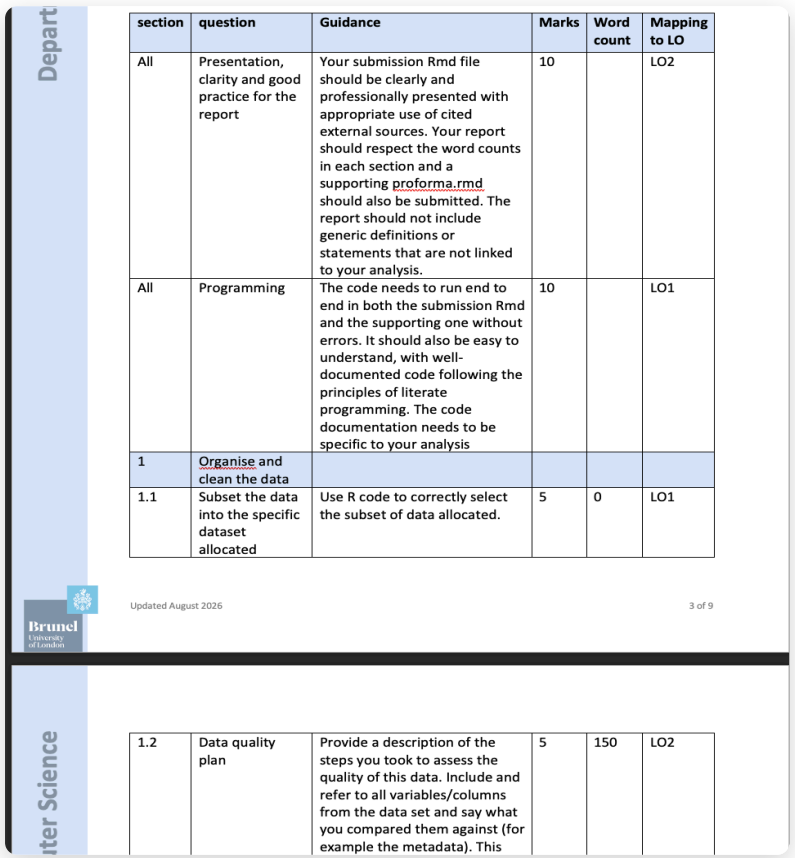
```r
# Only change the value for SID
# Assign your student id into the variable SID, for example:
SID <- 2101234                 # This is an example, replace 2101234 with your actual ID
SIDoffset <- (SID %% 50) + 1    # Your SID mod 50 + 1

load("health_club.Rda")
# Now subset the health club data set
# Pick every 50th observation starting from your offset
# Put into your data frame named mydf (you can rename it)
mydf <- df[seq(from=SIDoffset,to=nrow(df),by=50),]
```

Coursework – mark scheme

*(*see the complete list in the assessment brief)*

The mark scheme is detailed in the assessment brief document



section	question	Guidance	Marks	Word count	Mapping to LO
All	Presentation, clarity and good practice for the report	Your submission Rmd file should be clearly and professionally presented with appropriate use of cited external sources. Your report should respect the word counts in each section and a supporting proforma.rmd should also be submitted. The report should not include generic definitions or statements that are not linked to your analysis.	10		LO2
All	Programming	The code needs to run end to end in both the submission Rmd and the supporting one without errors. It should also be easy to understand, with well-documented code following the principles of literate programming. The code documentation needs to be specific to your analysis	10		LO1
1	Organise and clean the data				
1.1	Subset the data into the specific dataset allocated	Use R code to correctly select the subset of data allocated.	5	0	LO1
1.2	Data quality plan	Provide a description of the steps you took to assess the quality of this data. Include and refer to all variables/columns from the data set and say what you compared them against (for example the metadata). This	5	150	LO2

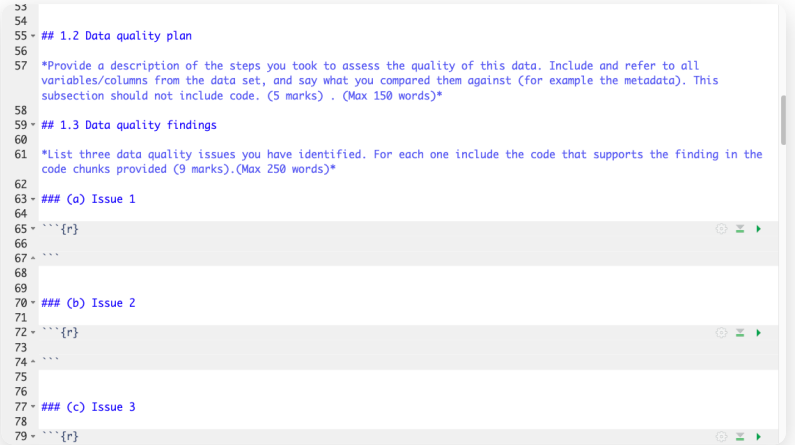
Coursework – guidance

*(*see the complete list in the assessment brief)*

- You are expected to use R and RMarkdown to complete the coursework
- Use `proforma-supporting-26-27.rmd` to conduct your investigation and analysis based on the questions asked
- Then use `proforma-submission-26-27.rmd` to answer the questions asked and summarise your investigations (where relevant)
- Keep the section headings
- Follow the principles of 'literate programming'

Coursework – Why two Rmd files?

- This is an extract of the `proforma-submission-26-27.rmd`
- The task in **1.2** requires you to describe steps taken, not to include the code for all the steps taken
- Use the `proforma-supporting.rmd` for the code for all the steps taken
- In task **1.3** you need to report on 3 issues in the `proforma-submission-26-27.rmd`
- Use the `proforma-supporting.rmd` to investigate your data so that you can report on the three issues found



```
## 1.2 Data quality plan

*Provide a description of the steps you took to assess the quality of this data. Include and refer to all
variables/columns from the data set, and say what you compared them against (for example the metadata). This
subsection should not include code. (5 marks) . (Max 150 words)*

## 1.3 Data quality findings

*List three data quality issues you have identified. For each one include the code that supports the finding in the
code chunks provided (9 marks).(Max 250 words)*

### (a) Issue 1

```{r}
```

(b) Issue 2

```{r}
```

(c) Issue 3

```{r}
```
```

Coursework – Submission files

*(*see the complete guidance in the assessment brief)*

In order to complete the assessment you will need to **submit two files**:

- A `proforma-submission-26-27.Rmd` file. This is the only file that will be used for marking, and needs to be clearly named (see submission instructions for details)
- A `proforma-supporting-26-27.Rmd` file. This file will also need to be submitted (see submission instructions for details)
- Both files need to run end to end

Coursework – File naming & the submission file

*(*see the complete guidance in the assessment brief)*

- Rename the files as explained in the Assessment Brief. For example, if your student id is [**EXAMPLE SID, e.g. 123456**] then: `123456-submission-26-27.Rmd` and `123456-supporting-26-27.Rmd`
- Consider the submission file as the summary report — e.g. if you have tried method X but it is not your final proposed model then it won't be in the `submission.rmd`, but you may mention trying it, and therefore it would be good practice to have what you tried in `supporting.rmd`

Coursework – guidance (2)

*(*see the complete list in the assessment brief)*

- The submission `rmd` file should be professionally presented with good structure, an absence of spelling errors and other typos, and written in an appropriate style (i.e., simple and to the point, unemotive language)
- Whilst we encourage collaboration and sharing of ideas, this is an individual report and so must be based on your own understanding, analysis and words
- **WiseFlow** automatically cross-compares all submissions

Coursework – guidance on GenAI use

*(*see the Expectation of AI use in the assessment brief)*

Generative AI (GenAI) and Large Language Models (LLMs) such as Copilot, chatGPT and Claude all have the potential to answer questions on code and material relevant to this assessment.

> **Important**
>
> Using AI-generated content and presenting it as your own original work in this assessment is **strictly prohibited**.

Despite many useful services — information search and retrieval, concept explanations, proofreading — it is crucial to recognise GenAI's inherent limitations:

- Responses can be overly general, inaccurate, biased, or even fabricated
- They often lack proper references and detailed insights
- Use may raise intellectual property and data privacy concerns

Coursework – guidance on GenAI use (cont.)

If you make use of Generative AI tools in any part of this assessment, it is essential to:

- **Acknowledge** that you have used GenAI, how you used it, and reference any AI content used in your submission (including code). If you need to reference or cite content from an AI tool, use the name of the tool used (e.g. Claude)
- **Avoid copying** explanations, text, or code directly from a GenAI tool into your submission. Use your own words to answer the questions asked. If you use GenAI to help with code, rigorously test and alter the code to fulfil the requirements of the assessment, and comment the code in your own words
- **Understand** any code you submit or explanation you provide, and be ready to explain it verbally. In the event of any concerns regarding the integrity of your work, an oral examination may be scheduled for further evaluation

> ⚠ **Warning**
>
> Misuse of Generative AI can lead to academic misconduct. See the link to the general University guidance above for more details.

Coursework – guidance: word count limit

*(*see the complete list in the assessment brief)*

- Make sure you respect the **1600 word limit** as well as the **specific section word limits**, as we discourage excessive padding
- Unnecessary words and waffle hinder and detract from professional presentation
- The word count does not include contents of code chunks or references
- Where answers to sections exceed the word limit for a section (see the marking criteria), words beyond the limit (with a discretionary 20% buffer) will be disregarded and you will lose marks

Coursework – guidance: word count limit (cont.)

*(*see the complete list in the assessment brief)*

- Including generic text will have two negative effects
- First, it will attract no credit. We already know that you need to check datasets for quality; rather you must be **specific** in terms of the actual data set you are asked to analyse
- Second, it wastes words from your word counts which we will strictly enforce
- So make sure your answer is couched in terms of variables and other specific qualities of your actual dataset. Where appropriate, cross-reference your answer to the relevant parts of the R code

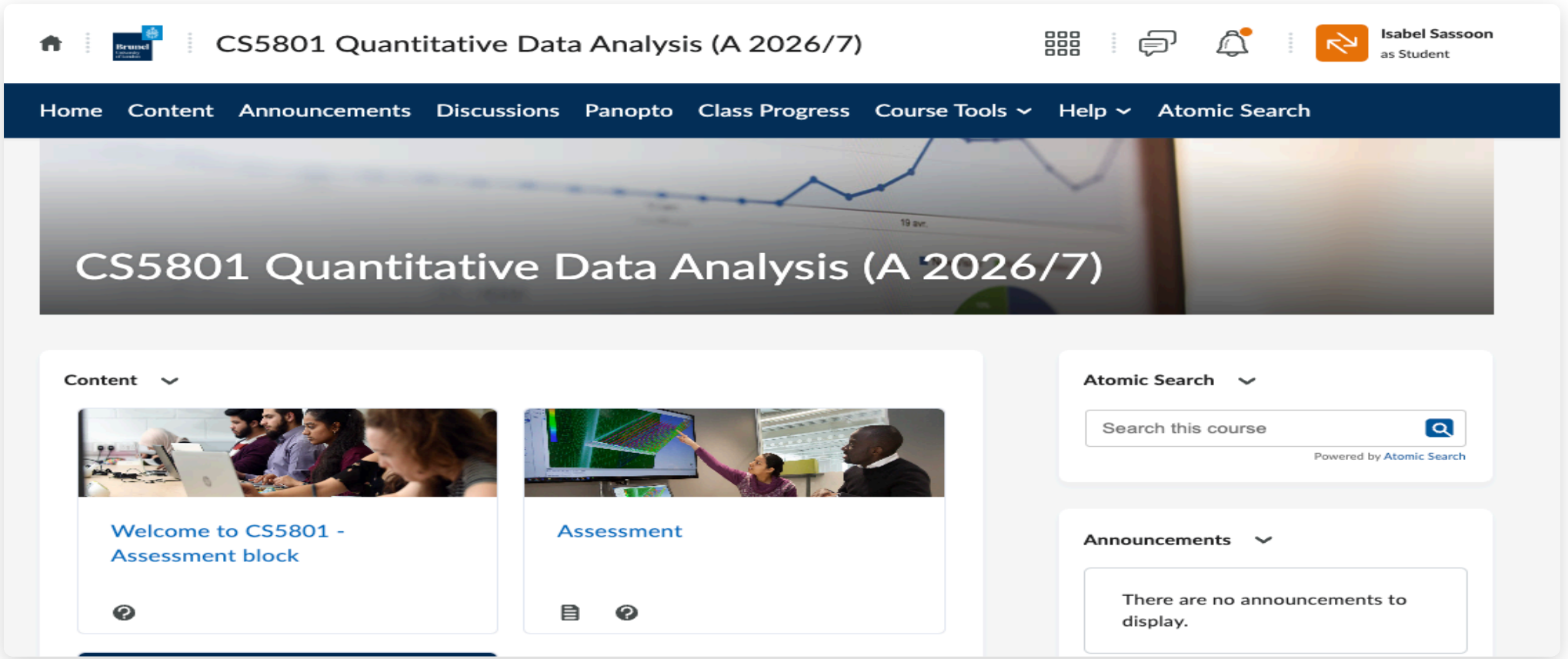
Coursework – a bad example

*(*see the complete list in the assessment brief)*

> "In order to analyze a dataset, one must first gather the relevant data and then study it. This involves looking at the numbers, making calculations, and drawing conclusions. It is important to examine the data closely and make sure it is accurate and relevant. By using various tools and techniques, one can derive meaningful insights from the data. Analysis is a critical step in the data processing cycle and should be approached with care and precision."

Although the above is true, it won't score any marks!

Coursework – Module page CS5801



Coursework – support

- We will allocate time during labs and also tutorials
- We are happy to help with general questions and will maintain an FAQ on Brightspace
- But…
  - We will not debug your code
  - We will not be offering a ‘pre-marking’ service, e.g. “Is this all I need do?”, “Is this good enough?”, “How can I get a better grade?”
  - Try to reframe such questions to: “Is it ok if I report on, and compare, two models for predicting [X]?”, “Should I include all the model diagnostics when I do [Y]?”
- There will be times during winter break when we’re all on leave

Coursework – end piece

- Please start early
- Give special attention to the data quality; do this first
- Read the brief very carefully (more than once)
- Add details and explanation
- If in doubt, ask us
- If in doubt, don’t trust the rumour mill!
- Remember we want you to do well

Coursework – Submitting

- You will submit your coursework through **WiseFlow**
- The link to submit will be made available at the end of the semester
- You submit two `.Rmd` files: a `1234567-proforma-submission-26-27.Rmd` file and a `1234567-proforma-supporting-26-27.Rmd` file (renamed using your student id)
- Other submission formats will **NOT** be graded and will be Fails
- We need to be able to run the code
- See the Assessment Brief and the Brightspace CS5801 page for all the files and information for this assessment

Coursework – Submission checklist

- ✅ YAML updated with name and student ID
- ✅ Libraries loaded at top of file
- ✅ Code runs end-to-end without errors
- ✅ Word count within 1600 words (excluding code/references)
- ✅ Remove guidance text and any install code
- ✅ Supporting file backs up claims in submission file
- ✅ Include references and bibliography (if relevant)
- ✅ Professional presentation: clear structure, no typos

Ask your questions on the discussion forum

This is set up on Brightspace CS5801 and there are already the answers from the questions asked during the launch session!